

TABLE 1
Number of errors committed when testing m null hypotheses

	Declared non-significant	Declared significant	Total
True null hypotheses	U	V	m_0
Non-true null hypotheses	T	S	$m - m_0$
	$m - R$	R	m

situation in a traditional form. The specific m hypotheses are assumed to be known in advance. R is an observable random variable; U , V , S and T are unobservable random variables. If each individual null hypothesis is tested separately at level α , then $R = R(\alpha)$ is increasing in α . We use the equivalent lower case letters for their realized values.

In terms of these random variables, the PCER is $E(V/m)$ and the FWER is $P(V \geq 1)$. Testing individually each hypothesis at level α guarantees that $E(V/m) \leq \alpha$. Testing individually each hypothesis at level α/m guarantees that $P(V \geq 1) \leq \alpha$.

2.1. Definition of False Discovery Rate

The proportion of errors committed by falsely rejecting null hypotheses can be viewed through the random variable $Q = V/(V + S)$ —the proportion of the rejected null hypotheses which are erroneously rejected. Naturally, we define $Q = 0$ when $V + S = 0$, as no error of false rejection can be committed. Q is an unobserved (unknown) random variable, as we do not know v or s , and thus $q = v/(v + s)$, even after experimentation and data analysis. We define the FDR Q_e to be the expectation of Q ,

$$Q_e = E(Q) = E\{V/(V + S)\} = E(V/R).$$

Two properties of this error rate are easily shown, yet are very important.

- If all null hypotheses are true, the FDR is equivalent to the FWER: in this case $s = 0$ and $v = r$, so if $v = 0$ then $Q = 0$, and if $v > 0$ then $Q = 1$, leading to $P(V \geq 1) = E(Q) = Q_e$. Therefore control of the FDR implies control of the FWER in the weak sense.
- When $m_0 < m$, the FDR is smaller than or equal to the FWER: in this case, if $v > 0$ then $v/r \leq 1$, leading to $\chi_{(V \geq 1)} \geq Q$. Taking expectations on both sides we obtain $P(V \geq 1) \geq Q_e$, and the two can be quite different. As a result, any procedure that controls the FWER also controls the FDR. However, if a procedure controls the FDR only, it can be less stringent, and a gain in power may be expected. In particular, the larger the number of the non-true null hypotheses is, the larger S tends to be, and so is the difference between the error rates. As a result, the potential for increase in power is larger when more of the hypotheses are non-true.

Third, consider $E(\mathbf{V})/E(\mathbf{R})$. When all hypotheses are true it is identically 1, and again impossible to control. A remedy may be given by either adding 1 to the denominator, a somewhat artificial solution, or by changing the denominator to $E(\mathbf{R}|\mathbf{R} > 0)$. Modifying both numerator and denominator in the same way will again run into problems of control when $m_0 = m$.

3. FALSE DISCOVERY RATE CONTROLLING PROCEDURE

3.1. The Procedure

Consider testing $H_1, H_2, \dots, H_m$ based on the corresponding p -values $P_1, P_2, \dots, P_m$. Let $P_{(1)} \leq P_{(2)} \leq \dots \leq P_{(m)}$ be the ordered p -values, and denote by $H_{(i)}$ the null hypothesis corresponding to $P_{(i)}$. Define the following Bonferroni-type multiple-testing procedure:

let k be the largest i for which $P_{(i)} \leq \frac{i}{m} q^*$;

then reject all $H_{(i)}$ $i = 1, 2, \dots, k$. (1)

Theorem 1. For independent test statistics and for any configuration of false null hypotheses, the above procedure controls the FDR at q^* .

Proof. The theorem follows from the following lemma, whose proof is given in Appendix A.

Lemma. For any $0 \leq m_0 \leq m$ independent p -values corresponding to true null hypotheses, and for any values that the $m_1 = m - m_0$ p -values corresponding to the false null hypotheses can take, the multiple-testing procedure defined by procedure (1) above satisfies the inequality

$$E(\mathbf{Q} | P_{m_0+1} = p_1, \dots, P_m = p_{m_1}) \leq \frac{m_0}{m} q^*. \quad (2)$$

Now, suppose that $m_1 = m - m_0$ of the hypotheses are false. Whatever the joint distribution of $P_1'', \dots, P_{m_1}''$ which corresponds to these false hypotheses is, integrating inequality (2) above we obtain

$$E(\mathbf{Q}) \leq \frac{m_0}{m} q^* \leq q^*,$$

and the FDR is controlled.

Remark. Note that the independence of the test statistics corresponding to the false null hypotheses is not needed for the proof of the theorem.

This procedure was mentioned by Simes (1986) as an exploratory extension to his procedure for rejection of the intersection hypothesis that all null hypotheses are true if, for some i , $P_{(i)} \leq i\alpha/m$. Whereas Simes (1986) showed that his procedure controls the FWER under the intersection null hypothesis, Hommel (1988) showed that the extended procedure for inference on individual hypotheses does not control the FWER in the strong sense: for some configuration of the false null hypotheses, the probability of an erroneous rejection is greater than α .

Hochberg (1988) has suggested a different way to utilize Simes's procedure so that it does control the FWER in the strong sense, by offering the following procedure:

let k be the largest i for which $P_{(i)} \leq \frac{i}{m+1-i} \alpha$;

then reject all $H_{(i)}$ $i = 1, 2, \dots, k$.

Note the relationship between Hochberg's procedure and the FDR controlling procedure when q^* is chosen to equal α . Both Hochberg's procedure and the FDR controlling procedure are step-down procedures, which start by comparing $p_{(m)}$ with α , and if smaller all hypotheses are rejected—as if a PCER approach had been taken. If $p_{(m)} > \alpha$, proceed to smaller p -values until one satisfies the condition. The procedures end, if not terminated earlier, by comparing $p_{(1)}$ with α/m , as in a pure Bonferroni comparison. At the two ends the procedures are similar, but, in between, the sequence of $p_{(i)}$ s is compared with $\{1 - (i - 1)/m\}\alpha$ in the current procedure, rather than with $\{1/(m + 1 - i)\}\alpha$ in Hochberg's procedure. The series of linearly decreasing constants of the FDR controlling method is always larger than the hyperbolically decreasing constants of Hochberg, and the extreme ratio is as large as $4m/(m + 1)^2$ at $i = (m + 1)/2$. This shows that the suggested procedure rejects samplewise at least as many hypotheses as Hochberg's method and therefore has also greater power than other FWER controlling methods such as Holm's (1979).

3.2. Example of False Discovery Rate Controlling Procedure

Thrombolysis with recombinant tissue-type plasminogen activator (rt-PA) and anisoylated plasminogen streptokinase activator (APSAC) in myocardial infarction has been proved to reduce mortality. Neuhaus *et al.* (1992) investigated the effects of a new front-loaded administration of rt-PA *versus* those obtained with a standard regimen of APSAC, in a randomized multicentre trial in 421 patients with acute myocardial infarction. Four families of hypotheses can be identified in the study:

- (a) base-line comparisons (11 hypotheses), where the problem is of showing equivalence;
- (b) patency of infarct-related artery (eight hypotheses);
- (c) reocclusion rates of patent infarct-related artery (six hypotheses);
- (d) cardiac and other events after the start of thrombolytic treatment (15 hypotheses).

In this last family FDR control may be desired: we do not wish to conclude that the front-loaded treatment is better if it is merely equivalent to the previous treatment in all respects.

In the paper, however, there is no attention to the problem of multiplicity (the only exception being the division of the end points into primary and secondary). The individual p -values are reported as they are, with no word of warning regarding their interpretation. The authors conclude that

'Compared to APSAC treatment, despite more early reocclusions, the clinical course with rt-PA treatment is more favorable with fewer bleeding complications and a substantially lower in-hospital mortality rate, presumably due to improved early patency of the infarct-related artery'.